2025 5th International Conference on Digital Futures and Transformative Technologies (ICoDT2) | 979-8-3315-5963-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICoDT269104.2025.11360748

A Multi-LLM Pipeline for Retrieval-Grounded, Bloom's Taxonomy-Aligned Question Generation

Muhammad Abdullah
National University of Sciences and Technology (NUST)
Islamabad, Pakistan
mabdullh.bscs22seecs@seecs.edu.pk

Isra Mansoor
National University of Sciences and Technology (NUST)
Islamabad, Pakistan
imansoor.bscs22seecs@seecs.edu.pk

Vitor Fortes Rey
Embedded Intelligence
German Research Center for Artificial Intelligence (DFKI)
Kaiserslautern, Germany
vitor.fortes@dfki.uni-kl.de

Muhammad Moazam Fraz
National University of Sciences and Technology (NUST)
Islamabad, Pakistan
moazam.fraz@seecs.edu.pk

***Abstract*—Automated assessment is becoming more common in education because it helps ease teachers' workload and makes evaluation more scalable. This doesn't exactly show how students are thinking, or what kind of knowledge they are struggling with. To tackle this problem, we built an agentic system containing a two-step framework that uses Large Language Models (LLMs) to generate and improve question and answer pairs for vocational education. In the first step, we used the Llama-2-7b model to create an initial set of 2288 question and answer pairs. In the second step, we used the more powerful Llama-3.3-70B-Instruct model. We applied Retrieval-Augmented Generation (RAG) to a knowledge base of textbooks and online resources to refine and expand these questions, resulting in a final corpus of 4292 items. The prompt for this enhancement is based on Bloom's taxonomy. It is used to map the difficulty levels of the question and answer pairs (easy: remember, understand; medium: apply, analyze; hard: evaluate, create). This enhanced Bloom's taxonomy framework also considers the type of knowledge involved: factual, conceptual, or procedural. Our results show that this framework not only scales up question generation but also makes the assessments more meaningful. It helps identify whether a student's difficulties come from memory, comprehension or deeper reasoning, giving educators a clearer picture of learning challenges.**

***Index Terms*—Automated assessment, Bloom's taxonomy, cognitive skills, educational technology, student evaluation.**

I. Introduction

Assessment plays an important role in education as it can be used to measure the achievement of students and provide feedback for improved learning. It does not only determine whether learning outcomes have been met but also guides instructors in tailoring instruction to address the learners' needs. Before the automation of assessments, traditional manual grading was used which was not only time consuming and subjective but often inconsistent across evaluators.

Recent advances in natural language processing and deep learning particularly with the rise of large language models, have enabled significant progress in the automated evaluation of objective, short answer and even essay type questions [1]. These systems promise efficiency and scalability, yet many existing approaches reduce assessment to a binary process in which answers are judged correct or incorrect [2]. For meaningful learning, it is critical to determine whether difficulties in learning arise from factual recall, conceptual understanding, procedural application or higher order reasoning. LLMs have the ability to process and analyze complex textual data, which they then use to offer a transformative opportunity to move beyond binary grading by providing context-aware feedback tailored to individual learner needs [3].

Bloom's taxonomy provides a well established cognitive framework for classifying educational objectives into hierarchical levels of complexity [4]. This enhanced version of Bloom's Taxonomy extends this framework by introducing two complementary dimensions which are Cognitive processes (remember, understand, apply, analyze, evaluate, and create) and Knowledge types (factual, conceptual, procedural, and metacognitive). This dual structure enables a better analysis of student performance as it moves beyond correctness to reveal both the depth of cognitive engagement and the type of knowledge being assessed. In the context of undergraduate education, the metacognitive dimension that involves reflection on one's own thinking and the creation of new knowledge is often beyond the scope of the study. However, the remaining dimensions offer a stronger basis for structured assessment. By aligning of automated assessments with this framework, educators' can better diagnose learning gaps and design targeted interventions [5].

In this paper, we propose an automated assessment system that integrates enhanced Bloom's taxonomy with LLM-based question generation and refinement. The system classifies assessment items not only by difficulty level but also by the type of knowledge they target. This allows instructors to identify specific weaknesses in students' learning. For example a student may demonstrate proficiency in factual recall but

This work is funded by DAAD (German Academic Exchange Service) project titled "Ba-Ikhtiyar Jawan: Upscaling & Digitisation of Vocational Education Curriculum in Pakistan"

Authorized licensed use limited to: Centro de Est. e Sistemas Avançados Recife (CESAR). Downloaded on June 30,2026 at 01:41:50 UTC from IEEE Xplore. Restrictions apply.

struggle with conceptual analysis. Similarly a student may succeed in procedural tasks yet falter in creative evaluation. By making use of these distinctions, the framework provides insights of the actions that instructors can take in order to design targeted feedback and engage learners' in self regulated improvement.

The following detail shows the organization of the different sections in this paper. Section II reviews the related work on automated assessment and the application of Bloom's taxonomy in educational technology. Section III describes the proposed methodology, including data collection, question generation and taxonomy integration. Section IV presents results and analysis of the system's performance and diagnostic capabilities. Finally Section V concludes with key findings, limitations and directions for future research.

II. Related Work

This section reviews prior research on the use of Large Language Models (LLMs) in education, focusing on their role in assessment, evaluation, and question generation. We organize the literature into five thematic strands: (1) LLMs for educational assessment, (2) LLMs as evaluators or "judges," (3) question generation guided by Bloom's taxonomy, (4) Bloom's taxonomy as a pedagogical foundation, and (5) foundational advances in large-scale language models.

A. LLMs for Educational Assessment

The first line of research explores how LLMs can be applied directly to student assessment in formal learning contexts. Agostini et al. [6] conducted an evaluation in higher education where LLMs were asked to grade authentic student assignments. Instead of relying solely on surface level correctness, these models were prompted using detailed rubrics and zero-shot instructions. These models then provide a score, an explanatory feedback and suggestions for improvement. The results demonstrated that LLMs can align with human raters. GPT-4o and Claude 3.5 showed the strongest reliability. However, the consistency of grading varied depending on the complexity of the task, highlighting both the potential and the limitations of these approaches. This shows the practical utility of LLMs in reducing the workload of instructors while also providing immediate and formative feedback to learners. Beyond efficiency, the integration of pedagogical rubrics into LLM prompts reflects a shift toward using AI. It is not merely as a grading assistant but also a tool for fostering learning outcomes.

B. LLMs as Evaluators ("LLM-as-a-Judge")

Zheng et al. [7] presented benchmarks like Chatbot Arena and MT-Bench. According to their results, GPT-4 attained 87–95% agreement with human evaluators which indicates that model can act as a substitute for knowledgeable judges in dialogue evaluation. Building on this, Chiang and Lee [8] investigated whether LLMs could serve as direct substitutes for human experts even in hostile environments. Their tests showed robustness across a range of inputs and confirmed a close alignment with human ratings. GPTScore being a probabilistic framework where LLMs act as judges across various tasks was introduced by Fu et al. [9]. GPTScore demonstrated the benefits of using semantic understanding instead of surface level overlaps by outperforming conventional automatic metrics like BLEU and ROUGE. ChatGPT was tested as a natural language generation evaluator by Wang et al. [10], who found strong correlations with human judgments in summarization and story generation. By using GPT-4 in information retrieval, Thomas et al. [11] demonstrated that the model could forecast user preferences using chain-of-thought reasoning and structured prompting. EvalGen, a framework that methodically creates evaluators in line with human preferences for tasks like sentiment analysis and named entity recognition, was proposed by Shankar et al. [12]. The paradigm was expanded by Verga et al. [13] through the introduction of "jury-based" evaluation, in which several LLMs collaborate as judges to reduce individual biases and produce more trustworthy consensus. Beyond scholarly contributions, Samuylova offered a useful summary of LLM as a judge techniques such as pairwise comparison, reference free, reference based scoring, and conversation level evaluation. These techniques are summed up in a straightforward workflow in which an LLM grader processes student assignments to produce feedback (Figure 3).

C. Question Generation and Bloom's Taxonomy

Parallel to evaluation research, another significant strand of work investigates LLMs for educational question generation, often guided by Bloom's taxonomy. Duong-Trung et al. [14] introduced BloomLLM, a system that combines supervised fine-tuning with Bloom's taxonomy to generate controlled questions across cognitive levels. Their results showed that explicitly embedding taxonomy-based constraints improved both the relevance and pedagogical value of generated questions.

Scaria et al. [15] evaluated the capability of modern LLMs to generate high-quality questions aligned with Bloom's categories in the Indian high school curriculum. They found that while models could reliably generate factual and comprehension-level items, higher-order questions involving application, analysis, and evaluation remained challenging. In follow-up work, Scaria et al. [16] proposed automated strategies for question generation aligned with Bloom's taxonomy, highlighting both opportunities and limitations in scaling these systems.

Other researchers explored related approaches. Al Faraby and Romadhony [17] demonstrated that LLMs can adaptively generate and classify questions across Bloom's levels, underscoring their versatility in educational contexts. Li and Zhang [18] proposed a "planning-first, question-second" framework, where LLMs generate a structured outline before formulating questions, thereby improving coherence and alignment with intended cognitive skills. Rao et al. [19] provided a comprehensive review of automatic Q&A generation methods, summarizing trends, datasets, and challenges. Mucciaccia et al. [20] went further by implementing an LLM-based system

Authorized licensed use limited to: Centro de Est. e Sistemas Avançados Recife (CESAR). Downloaded on June 30,2026 at 01:41:50 UTC from IEEE Xplore. Restrictions apply.

for generating and evaluating multiple-choice questions in university-level materials.

D. Bloom's Taxonomy as a Pedagogical Foundation

The foundation of this study lies mainly on Bloom's taxonomy and it's later versions. Wilson [21] reviewed Anderson and Krathwohl's revised taxonomy in which orignal Bloom's hierarchy was updated by distinguishing between knowledge dimensions and cognitive processes. This enhanced framework provides a more sophisticated understanding that how learning outcomes can be structured and assessed. The taxonomy specifically focus on progressive levels of cognitive engagement that makes it suitable for guiding LLM-based question generation and assessment design. By incorporating these levels into model prompts it can be ensured that automated systems capture not only factual recall but also higher-order thinking skills.

E. Foundational Advances in Large Language Models

The advancements in large scale pre-trained modeks has increased the applications of LLMs in automated assessments and pedagogy. Brown et al. [22] demonstrated that GPT-3 can perform a wide range of tasks in a few-shot or zero-shot setting which revealed the emergent capabilities of scaling. Studies have validated that few-shot prompting with long-context models, can effectively resolve complex linguistic dependencies without the need for task-specific fine-tuning [23]. These findings catalyzed a wave of research into educational applications, as researchers began to test that general-purpose models could be adapted for structured, pedagogically meaningful tasks.

Further advancements in many models have improved reasoning, factual accuracy and controllability which makes them more suitable for high stakes fields like education.

In summary, prior research demonstrates that LLMs can act as evaluators of student tasks, chatbot outputs and information retrieval results, often achieving high alignment with human judges. In parallel, LLMs are increasingly applied to question generation tasks guided by Bloom's taxonomy. Together, these two strands LLMs as evaluators and LLMs as pedagogical designers form the basis for our study, which leverages an enhanced Bloom's taxonomy for automated educational assessment.

III. Proposed Methodology

The proposed methodology builds on the use of a two-stage large language model (LLM) pipeline that combines domain-specific data, retrieval-augmented generation (RAG), and the integration of Enhanced Bloom's Taxonomy to generate high-quality assessment items. The overall goal of this approach is to automate the creation of structured questions and answers for the *Auto-Electrician* course, while also enabling diagnostic insights into students' cognitive and knowledge-based abilities.

A. Stage I: Data Collection and Preprocessing

In the first stage, we collected a total of **656** YouTube video transcripts related to the *Auto-Electrician* course. This field was chosen carefully because it has strong balance between theoretical elements such as electrical principles, wiring diagrams, safety procedures and practical aspects such as diagnosing, testing and repairing automotive electrical components. YouTube videos in contrast to traditional textbooks offer vivid examples of practical applications, making them particularly suitable for vocational program such as auto-electrician training.

From overall dataset, a subset of **262** transcripts was selected for initial processing. These transcripts were divided into smaller, semantically coherent chunks in order to be easily given to LLMs. This preprocessing step was necessary because long transcripts often exceed the input length limitations of LLMs and unsegmented content may reduce the quality of generated outputs. The resulting dataset ensured that both theoretical explanations and practical demonstrations were captured in content.

B. Stage II: Question Generation Using the Basic LLM

The first stage of question generation employed the, **meta-llama/Llama-2-7b-chat-hf** model, which we designated as the baseline LLM in our pipeline. Following decoding settings were used while generation process: we used nucleus sampling with top_p = 0.9 and 0.6 was the temperature. New tokens was set to 1024 and input transcripts were truncated at 4096 tokens This model was tasked with creating question–answer (QA) pairs directly from the preprocessed YouTube transcripts. The model generated questions across three levels of difficulty: easy, medium, and hard. This categorization helped in two ways. On one hand, it helps us to understand the learners level of understanding. For example, a student who struggles with easy questions is likely at the beginner stage and must strengthen foundational knowledge before advancing to more challenging tasks. On the other hand, it helps us to provide a tailored assessment pathway according to performance of student . In this first stage, the LLaMA-13B model generated 2288 QA pairs from the 262 transcripts.

C. Stage III: Refinement with Enhanced LLM and RAG

In the second stage, model filtered out low-quality questions and restructured them using Enhanced Bloom's Taxonomy for pedagogical alignment. For this purpose, we employed the **meta-llama/Llama-3.3-70B-Instruct** model, a more advanced LLM which was loaded using 4-bit quantization (load_in_4bit = True) with a torch.bfloat16 data type to provide greater capacity and reasoning ability. A retrieval-augmented generation (RAG) framework was included, which incorporated reliable domain specific resources to outputs.

The RAG component included web-scraped technical blogs(e.g 2CarPros.com, HumbleMechanic.com), service manuals, websites, and multiple textbooks on auto-electrician training. All materials were used for research purposes. The content was parsed, chunked and then embedded using the

Authorized licensed use limited to: Centro de Est. e Sistemas Avançados Recife (CESAR). Downloaded on June 30,2026 at 01:41:50 UTC from IEEE Xplore. Restrictions apply.

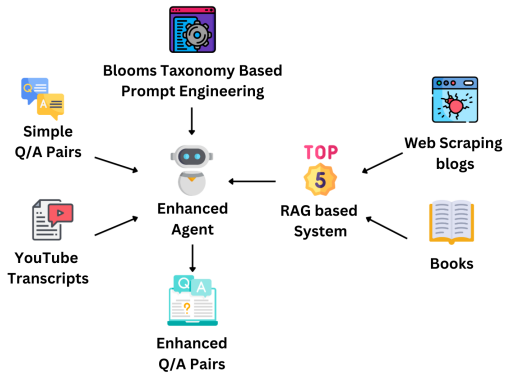


Fig. 1. Workflow of the enhanced agent: input from books, blogs, and initial Q&A pairs is refined using Bloom's Taxonomy–based prompt engineering to generate pedagogically enhanced questions.

Remember	Understand	Apply	Analyze	Evaluate	Create
Define	Summarize	Solve	Organize	Criticize	Design
Identify	Compare	Change	Contrast	Justify	Modify
Copy	Interpret	Relate	Connect	Judge	Develop
State	Classify	Complete	Relate	Defend	Rewrite
List	Discuss	Use	Devise	Appraise	Invent
Describe	Categorize	Sketch	Correlate	Assess	Write
Recognize	Contrast	Teach	Illustrate	Prioritize	Formulate
Find	Infer	Articulate	Distill	Plan	Compose
Tell	Report	Discover	Conclude	Grade	Construct
Explain	Relate	Transfer	Take Apart	Reframe	Simulate
Recite	Extract	Determine	Deduce	Value	Build
Memorize	Translate	Implement	Link	Argue	Collaborate
Illustrate	Paraphrase	Reenact	Test	Reflect	Pivot
Quote	Cite	Present	Experiment	Support	Blog

Fig. 2. Action verbs of Enhanced Bloom's Taxonomy categorized across six cognitive levels

all-MiniLM-L6-v2 SentenceTransformer model. These vector embeddings were indexed and stored in a vector database. For each transcript, query was executed against index to retrieve the top-k=3 most relevant text chunks. No re-ranker was used in this pipeline. These chunks were then added to prompt as context. Notably, the textbooks included both English resources and translated German materials, broadening the coverage of theoretical and procedural content. This multi source integration helped to refine questions which were not only based on Youtube transcripts but also informed by credible educational references. The generation parameters were max_new_tokens=2048 while the temperature and top_p values remained 0.6 and 0.9 respectively. Figure 1 depicts how raw QA pairs from transcripts are transformed into enhanced items using books and blogs.

D. Stage IV: Integration of Enhanced Bloom's Taxonomy

The addition of Enhanced Bloom's taxanomy is the key novelty of the the methodology. Cognitive depth and knowledge type were emphasized by this framework instead of relying on binary correctness as in traditional assessment methods. The enhancement introduces two dimensions: (1) cognitive processes which include levels of remembering, understanding, applying, analyzing, evaluating, and creating and (2) knowledge types which are factual, conceptual, procedural, and metacognitive. Instructors can determine the type and degree of students' knowledge more accurately along with the accuracy of their responses due to this combination of characteristics. This method supports more personalized feedback and instructional design which helps learners strengthen their cognitive skills and knowledge areas.

We matched the difficulty levels produced in Stage II to certain cognitive categories in order to fully integrate this taxonomy within our system. Easy-level questions were included in category of remembering and understanding. They were restructured with verbs such as *define, identify, describe, summarize, and interpret* to represent the expected cognitive demand. Applying and analyzing of taxonomy covered medium level questions which used verbs like *solve, implement, organize, and experiment* and it tests student's ability to apply knowledge and break it into components. Hard-level questions were mapped with evaluating and creating where verbs like *criticize, justify, design, and compose* were used which emphasize on judgment, synthesis and innovation. Figure 2 shows this alignment and distribution of action verbs throughout Bloom's cognitive categories.

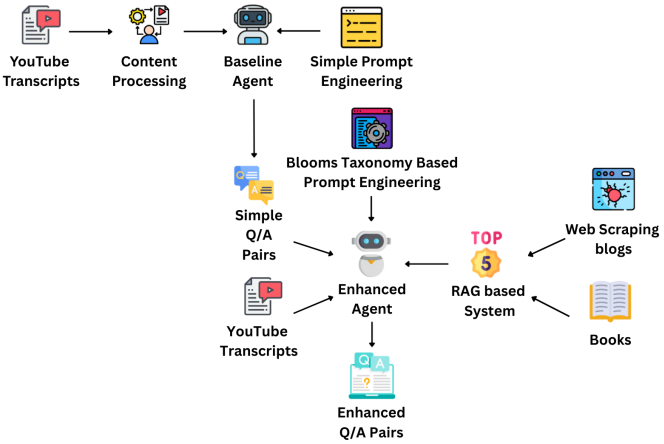


Fig. 3. Overview of the proposed system workflow: Base LLM process the raw transcripts to generate initial Q&A pairs which are refined by another LLM using RAG and Bloom's Taxonomy to produce structured, pedagogically aligned enhanced Q&A pairs.

In additon to improving the clarity of questions this enhancement ensured that each is connected to a cognitive and knowledge dimension. As a result, instructors can diagnose whether a student's weakness lies in recalling factual knowledge, conceptual understanding or procedural application which is way better than simply marking correct or incorrect answers. The metacognitive dimension was excluded from this work. In a vocational program like auto-electrician training, it is neither feasible nor reliable to assess whether students can propose entirely new strategies or techniques and they are correct or not. For example, if a student suggests a new repair method. It would be challenging for the system to

Authorized licensed use limited to: Centro de Est. e Sistemas Avançados Recife (CESAR). Downloaded on June 30,2026 at 01:41:50 UTC from IEEE Xplore. Restrictions apply.

determine its correctness so this dimension was removed as a limitation of the current work.

The LLaMA-70B model expanded the initial 2,288 QA pairs into a more diverse and enhanced set of **4,292** questions. This significant increase reflects a broader coverage of the subject matter and demonstrates the value added by Bloom's-aligned enhancements.

E. Stage V: Diagnostic Insights and Limitations

The proposed approach generates assessments along with diagnostic functions. This method can provide detailed feedback on students' performance by mapping assessment questions to both Bloom's dimensions and difficulty levels. For example, a student may show strong factual knowledge but he can struggle with conceptual understanding. Alternatively, a learner may be successful in applying knowledge but fail when asked to evaluate or create. This diagnostic precision helps instructors to design more targeted treatments and enables learners to engage in regulated improvement.

But this approach also has significant drawbacks as well. The most prominent is the removal of the metacognitive dimension of Enhanced Bloom's taxonomy. Although metacognition is a higher order of thinking that involves the creation of novel ideas. It is not feasible to assess these abilities automatically in the context of vocational training. More over, using YouTube transcripts as the primary data source although rich in practical content but it also creates potential inconsistencies in terminology and instructional depth. These limitations highlight possible opportunities for future work that include integration of expert-curated questions or developing mechanisms to evaluate metacognitive skills in applied contexts.

F. Workflow Overview

The overall workflow of proposed system is displayed in Figure 3. Raw transcripts are processed by a base LLM to extract and generate base preliminary Q&A pairs in the first stage which serves as a foundation of assessment pipeline. The initial pairs are refined by an enhanced LLM which incorporates retrieval-augmented generation (RAGs) to enrich the questions with external sources such as textbooks, blogs, and technical documents which improves the accuracy, reliability and domain coverage. To guarantee pedagogical validity, the refinement process integrates Enhanced Bloom's Taxonomy, which directs the restructuring of questions along cognitive processes and knowledge types. As a result, the pipeline allows for evaluations that are both scalable and diagnostically significant by classifying questions according to their level of difficulty as well as determining the cognitive and knowledge dimension they belong to. This ensures that the system can provide large scale evaluations that are both scalable and pedagogical aligned while still offering actionable feedback for learners and instructors.

IV. Results

To show the how well and effectively the proposed refinement pipeline works, we can make a comparison between initial Q&A pairs generated by the base LLM and their enhanced counterparts generated after the integration of retrieval-augmented generation (RAGs) and Enhanced Bloom's Taxonomy. The following examples display how simple, raw and straightforward questions are systematically transformed into more comprehensive, pedagogically aligned pairs that not only increase clarity and depth but also incorporate difficulty levels and knowledge dimensions.

Simple Q&A Example 1:

Q: What is the purpose of using an adhesion promoter on a painted surface? Difficulty: Easy

A: Using an adhesion promoter on a painted surface can help to improve the bonding of paint to non-prepared surfaces, such as glass or chrome.

Enhanced Q&A Example 1:

Q: Explain the purpose of using an adhesion promoter when painting a brand new, clean alternator, and summarize the benefits of this product in ensuring paint adhesion to non-prepared surfaces.

A: The adhesion promoter is used to help paint stick to non-prepared surfaces, such as shiny metals. This product is beneficial in situations where the surface is not prepared for painting, as it ensures a strong bond between the paint and the surface, even on materials like glass and chrome.

Difficulty: Easy Knowledge Dimension: Factual

Simple Q&A Example 2:

Q: How does the switch in the video turn on the relays for the turn signals and brake lights?

A: The switch in the video sends power through the white wires to the center of the switch, which then pushes power out to the two wires connected to pin 86 on the relays, turning the turn signals on. When the brake lights are connected, power is sent through the fuse box and red wires into the brake switch, which then turns on the brake lights through pin 87a of the rear light relays.

Difficulty: Intermediate

Enhanced Q&A Example 2:

Q: Analyze the brake light wiring and determine how the brake switch and relay interact to control the brake lights and turn signals.

A:The brake lights use pin 87a on the relays, with power coming from the fuse box through the red wires into the brake switch. The brake switch sends power to pin 87a, which is normally connected to pin 30. However, when the relay is turned on, pin 87a is disconnected from pin 30, and pin 87 (the yellow flasher wire) is connected to pin 30, effectively turning the brake lights into turn signal lights.

Difficulty: Intermediate Knowledge Dimension: Procedural

Authorized licensed use limited to: Centro de Est. e Sistemas Avançados Recife (CESAR). Downloaded on June 30,2026 at 01:41:50 UTC from IEEE Xplore. Restrictions apply.

The integration of retrieval-augmented generation has ensured that the content is more diverse and grounded with reliable sources. The use of Enhanced Bloom's Taxonomy has promoted high order thinking by aligning questions with cognitive processes and knowledge dimensions. The enhanced question answer pairs are rich in detail and they are more pedagogically structured and better suited for diagnostic assessment as compared to the simple Q&A pairs which makes them more valuable for educational contexts.

V. Conclusion

This paper presents a novel and unique framework for autmated assessment where LLM integrates with RAGs using Enhanced Bloom's Taxanomy. In contrast to traditional systems that mainly evaluate correctness, this pipeline generates structured, pedagogically aligned question answers that distinguish between cognitive functions and knowledge dimensions. Findings of our method have shown that it produces scalable and diagnostically meaningful assessments. It provides practical information and insights of learners' strengths and weaknesses. This framework has established a solid foundation for future work but reliance on YouTube transcripts and exclusion of the meta-cognitive dimension represents current limitations. Broader range of data sources, adaptive feedback mechanisms and large scale validation with student performance data to support pedagogical effectiveness of assessments can extend this scope. Overall, the study shows the potential of combining LLMs with well established educational theory to support both automated grading and deeper learning outcomes in a scalable manner.

VI. Data and Code Availability

Complete data set containing all 2,288 initial and 4,292 enhanced QA pairs, prompt templates and implementation details are available upon reasonable request.

References

[1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, 2019, pp. 4171–4186.

[2] R. E. Bennett, "Formative assessment: A critical review," *Assessment in education: principles, policy & practice*, vol. 18, no. 1, pp. 5–25, 2011.

[3] E. Kasneci, K. Seßler, S. Küchemann, M. Bannert, D. Dementieva, F. Fischer, U. Gasser, G. Groh, S. Günnemann, E. Hüllermeier *et al.*, "Chatgpt for good? on opportunities and challenges of large language models for education," *Learning and individual differences*, vol. 103, p. 102274, 2023.

[4] D. R. Krathwohl, B. S. Bloom, and B. B. Masia, "Taxonomy of educational objectives: The classification of educational goals; handbook," 1966.

[5] D. R. Krathwohl, "A revision of bloom's taxonomy: An overview theory into practice, 41: 4, 212-218," *DOI*, vol. 10, p. s15430421tip4104_2, 2002.

[6] D. Agostini, F. Picasso, H. Ballardini *et al.*, "Large language models for the assessment of students' authentic tasks: A replication study in higher education," in *CEUR Workshop Proceedings*, vol. 3879, 2024.

[7] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. Xing *et al.*, "Judging llm-as-a-judge with mt-bench and chatbot arena," *Advances in neural information processing systems*, vol. 36, pp. 46 595–46 623, 2023.

[8] C.-H. Chiang and H.-y. Lee, "Can large language models be an alternative to human evaluations?" *arXiv preprint arXiv:2305.01937*, 2023.

[9] J. Fu, S.-K. Ng, Z. Jiang, and P. Liu, "Gptscore: Evaluate as you desire," *arXiv preprint arXiv:2302.04166*, 2023.

[10] J. Wang, Y. Liang, F. Meng, Z. Sun, H. Shi, Z. Li, J. Xu, J. Qu, and J. Zhou, "Is chatgpt a good nlg evaluator? a preliminary study," *arXiv preprint arXiv:2303.04048*, 2023.

[11] P. Thomas, S. Spielman, N. Craswell, and B. Mitra, "Large language models can accurately predict searcher preferences," in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024, pp. 1930–1940.

[12] S. Shankar, J. Zamfirescu-Pereira, B. Hartmann, A. Parameswaran, and I. Arawjo, "Who validates the validators? aligning llm-assisted evaluation of llm outputs with human preferences," in *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, 2024, pp. 1–14.

[13] P. Verga, S. Hofstatter, S. Althammer, Y. Su, A. Piktus, A. Arkhangorodsky, M. Xu, N. White, and P. Lewis, "Replacing judges with juries: Evaluating llm generations with a panel of diverse models," *arXiv preprint arXiv:2404.18796*, 2024.

[14] N. Duong-Trung, X. Wang, and M. Kravčík, "Bloomllm: Large language models based question generation combining supervised fine-tuning and bloom's taxonomy," in *European Conference on Technology Enhanced Learning*. Springer, 2024, pp. 93–98.

[15] N. Scaria, S. D. Chenna, and D. Subramani, "How good are modern llms in generating relevant and high-quality questions at different bloom's skill levels for indian high school social science curriculum?" in *Proceedings of the 19th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2024)*, 2024, pp. 1–10.

[16] N. Scaria, S. Dharani Chenna, and D. Subramani, "Automated educational question generation at different bloom's skill levels using large language models: Strategies and evaluation," in *International Conference on Artificial Intelligence in Education*. Springer, 2024, pp. 165–179.

[17] S. Al Faraby, A. Romadhony *et al.*, "Analysis of llms for educational question classification and generation," *Computers and Education: Artificial Intelligence*, vol. 7, p. 100298, 2024.

[18] K. Li and Y. Zhang, "Planning first, question second: An llm-guided method for controllable question generation," in *Findings of the Association for Computational Linguistics ACL 2024*, 2024, pp. 4715–4729.

[19] M. M. Rao, N. Bharath, R. Reddy, S. Raviteja, and V. Ankitha, "Review paper on automatic q&a generator using llm (large language model)," *NBV and Reddy, RV Praneeth Siddhartha and Raviteja, S and Ankitha, Vadiyala, Review Paper on Automatic Q&A Generator Using LLM (Large Language Model)(May 2, 2024)*, 2024.

[20] S. S. Mucciaccia, T. M. Paixão, F. W. Mutz, C. S. Badue, A. F. de Souza, and T. Oliveira-Santos, "Automatic multiple-choice question generation and evaluation systems based on llm: A study case with university resolutions," in *Proceedings of the 31st International Conference on Computational Linguistics*, 2025, pp. 2246–2260.

[21] L. O. Wilson, "Anderson and krathwohl–bloom's taxonomy revised," *Understanding the new version of Bloom's taxonomy*, 2016.

[22] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, "Language models are few-shot learners," *Advances in neural information processing systems*, vol. 33, pp. 1877–1901, 2020.

[23] M. Sajid, M. Fraz, S. Latif, and Z. Zafar, "Few-shot multilingual coreference resolution using long-context large language models," in *Proceedings of the Eighth Workshop on Computational Models of Reference, Anaphora and Coreference*, 2025, pp. 154–162.

Authorized licensed use limited to: Centro de Est. e Sistemas Avançados Recife (CESAR). Downloaded on June 30,2026 at 01:41:50 UTC from IEEE Xplore. Restrictions apply.